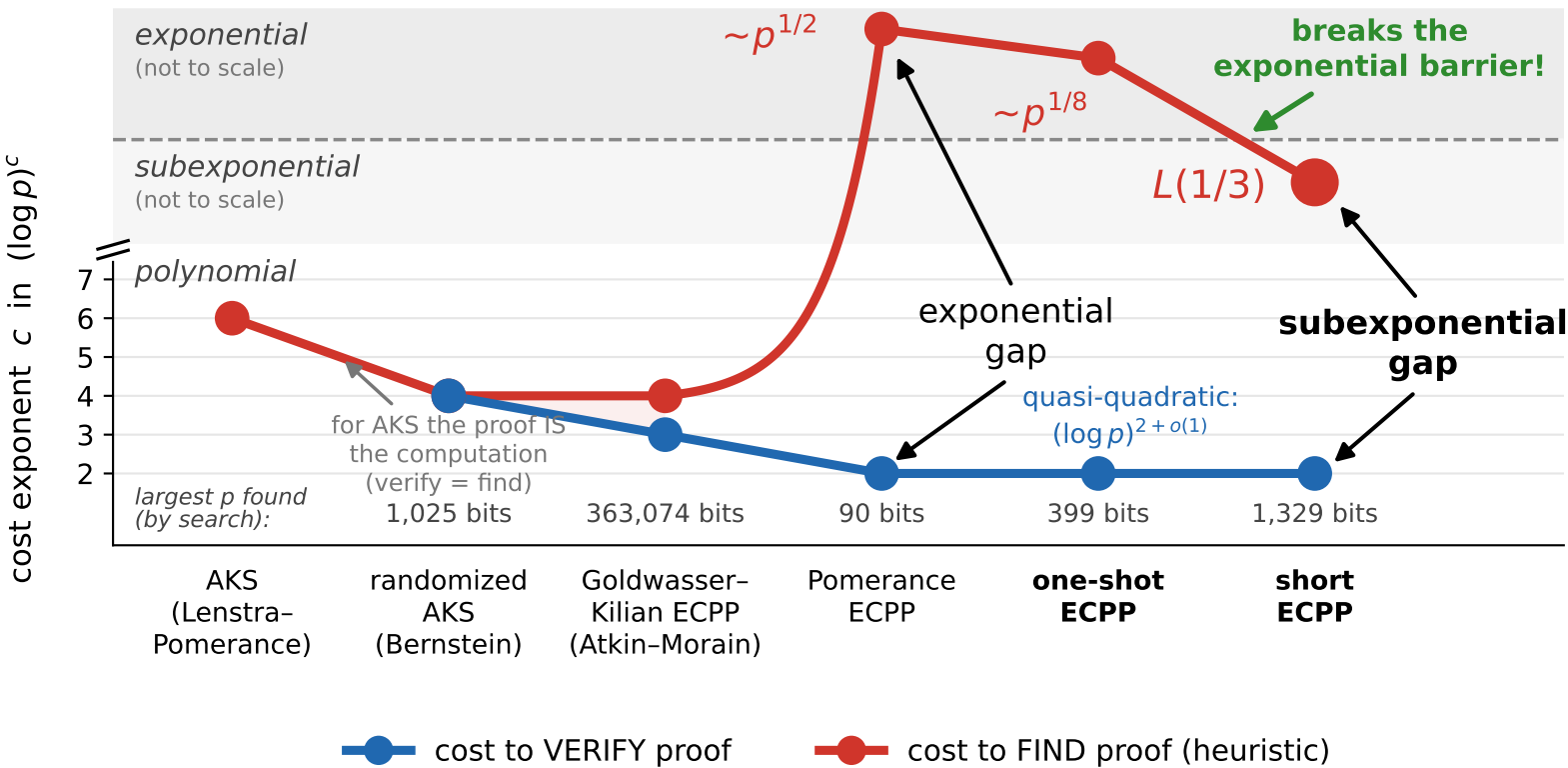
exponential
(not to scale)
subexponential
(not to scale)
polynomial
cost exponent c in $(\log p)^c$
7
6
5
4
3
2
$\sim p^{1/2}$
$\sim p^{1/8}$
breaks the exponential barrier!
$L(1/3)$
exponential gap
subexponential gap
quasi-quadratic: $(\log p)^{2+o(1)}$
for AKS the proof IS the computation (verify = find)
largest p found (by search):
1,025 bits
363,074 bits
90 bits
399 bits
1,329 bits
AKS (Lenstra–Pomerance)
randomized AKS (Bernstein)
Goldwasser–Kilian ECPP (Atkin–Morain)
Pomerance ECPP
one-shot ECPP
short ECPP
cost to VERIFY proof
cost to FIND proof (heuristic)